Joseph Yanney-Ewusie Newton

Bonn, Germany
+49 176 85623421 • jyenewton10@gmail.com
LinkedIn, GitHub, Portfolio

Climate & Agri-Food Data | GHG Accounting ▪ Carbon MRV ▪ Geospatial

MSc in Environmental Sustainability with a BSc in Agricultural Engineering, focused on climate and agri-food data. Builds data tools and pipelines for emissions, land-use and farm data. Draws on hands-on experience in Ghana, Israel and Germany to turn messy field and supply-chain data into practical answers for sustainability decisions.

Core Competencies

- **Carbon & Compliance**: GHG Protocol Scope 1 & 2, IPCC emission factors, carbon MRV, REDD+ / Verra (VM0048), VSME/ESRS mapping, Monte Carlo uncertainty analysis, Net Zero frameworks, Nature-Based Solutions.
- **Geospatial & Remote Sensing**: Google Earth Engine, Sentinel-2, Landsat, land-cover classification (Random Forest), change detection, QGIS, raster and vector analysis (GeoTIFF, GeoJSON).
- **Programming & Data**: Python (pandas, numpy, rasterio), SQL, PostgreSQL, ETL pipelines, time-series analysis, data quality validation and documentation.
- **Reporting & Visualization**: Power BI, Tableau, Streamlit, Excel, Matplotlib.
- **Tools & Collaboration**: Git/GitHub, AI-assisted development with Claude Code, Jira, Confluence, Microsoft 365, PowerPoint.

Selected Projects

- **Forest Carbon MRV Prototype, Kakum National Park, Ghana**
  GitHub
  - Built a forest carbon MRV pipeline structured to Verra REDD+ (VM0048), estimating about 12.5M tCO_2e (about 4.8M conservative, ±61.5%) from Sentinel-2, using Random Forest classification in Google Earth Engine (95.5% accuracy) and a tested Python package.
  - Identified the biomass default, not the classification, as the main source of error, and prioritized a GEDI-based Tier 2 biomass upgrade.
- **GHG Inventory Tool (Scope 1 & 2), personal project**
  Live demo, GitHub
  - Built a Python GHG Protocol inventory with per-gas calculations, dual Scope 2 reporting, Monte Carlo uncertainty, VSME/ESRS mapping, and audit-ready PDF and Excel outputs.
  - Added an abatement analysis that ranks seven reduction options by € per tonne, and deployed it as a Streamlit app for data editing, scenarios and report export.
- **Green Roof Climate Data Pipeline and Dashboard, MSc thesis**
  GitHub
  - Built an ETL pipeline that QC-filters and harmonizes multi-vendor sensor data into a 2.18M-record time-series warehouse (2020-2026), with a Streamlit dashboard for significance testing and energy-balance analysis.

Professional Experience

- **HTWest GmbH** — **Bonn, Germany**
  Werkstudent, Production Assistant — *Apr 2026-Present*
  - Support production, packaging and shipping of precision-cut specialty papers and films, including Parafilm, filter media and wrappers, for B2B customers.
  - Check orders, labels and finished products.

- **BASF Digital Farming GmbH (xarvio)** — **Cologne, Germany**
  Werkstudent, Data Quality (Drone Data & Sustainable Weed Management) — *Oct 2024-Feb 2026*
  - Validated and documented drone datasets for weed-detection models, supporting reliable model outputs and operational decisions.
  - Coordinated with teams in Germany, Argentina and Brazil to resolve data quality issues and align on process standards.
  - Wrote stakeholder summaries and status updates, tracked recurring quality issues and supported corrective actions.
- **Technische Hochschule Bingen** — **Bingen am Rhein, Germany**
  University Research Assistant — *Sep 2023-Dec 2023*
  - Collected, validated and structured field sensor data for climate adaptation research on crop moisture stress and temperature impacts.
  - Kept data traceable across monitoring locations through consistent documentation and quality checks.
- **Freezelink Cold Chain Logistics** — **Accra, Ghana**
  Project Analyst — *Jan 2022-Aug 2022*
  - Compiled benchmark reports on cold chain performance across client sites, used to guide implementation and operational improvement decisions.
  - Supported the planning and rollout of modular cooling solutions, coordinating information between project teams.
  - Maintained structured project documentation and produced efficiency reports for client reviews.
- **Tzafarti Ibudim (Citrus)** — **Moshav Ahituv, Israel**
  Trainee — *Dec 2020-Oct 2021*
  - Led a multicultural team of 10+ people during peak harvest, supporting a 95% product acceptance rate through structured quality oversight.
  - Managed irrigation systems on 50+ hectares of citrus.
- **Econexus Ventures Limited** — **Accra, Ghana**
  Production Team Lead — *Jan 2019-Oct 2020*
  - Led a cross-functional R&D team that took Ecogel, a bioethanol clean-cooking fuel, from formulation to market-ready production.

Education

- **Technische Hochschule Bingen** — **Germany**
  MSc, Environmental Sustainability
  Thesis: "Green Roof Cooling Effectiveness Analysis" (urban climate adaptation, environmental data pipeline design).
- **University of Ghana** — **Accra, Ghana**
  BSc, Agricultural Engineering
- **AgroStudies Training Center** — **Netanya, Israel**
  Diploma in Agriculture (Horticulture)

Volunteering

- **Volunteer Data Analyst**, Net Zero Tracker (since 2023): documents the net zero commitments of 50+ companies and countries, producing structured summaries for sustainability reporting.
- **Environmental Volunteer**, Service Civil International (SCI Germany), Upper Middle Rhine Valley: landscape restoration and erosion control in vineyard terrain.

Certifications

- OPF Academy (Cohort 6): Net Zero Strategies, GHG Accounting, Sustainability Frameworks ▪ Google Data Analytics Professional Certificate, Coursera ▪ BCG Climate & Sustainability Job Simulation, Forage ▪ Green Digital Skills, INCO Academy

Languages

- English (fluent); German (A2, actively learning).

Publications

- Panferov, O., Hietel, E., Rössner, U., Alef, J., & **Newton, J.** (2026). Rainwater-irrigated greenroofs as climate change adaptation in urban areas. EGU General Assembly 2026, BG10.12. doi:10.5194/egusphere-egu26-18880